



Paediatric Diabetic Ketoacidosis

Fluid and Insulin Prescription Sheet

- Use this documentation for children and young people aged under 18 years with DKA.
- This document is designed to be used with the MSF Diabetes Calculator.

Name	Date of birth	Medical record number/ID
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 Audit ID	Use the MSF Diabetes Calculator: (diabetes.msf.net) and record the Audit ID shown on the guidance page here. Enter other values from the guidance page below. If no infusion pump available, also record the infusion set drop factor and for each rate the drops/minute.
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 DKA severity	
Standard / Severe	7.5% fluid deficit / 10% fluid deficit
(delete as appropriate)	

 Fluid bolus
Administer a fluid bolus of Ringer lactate (or sodium chloride 0.9%) IV then reassess after.
mL over minutes at mL/hour

 Fluid replacement		
See preparation of IV fluids for two-bag method guidance overleaf.		
Select starting rate by checking blood glucose when IV fluids for two-bag method have been made up. Adjust according to blood glucose checks at least once per hour.		
Blood glucose level	Which bag(s) at what speed?	
	Bag 1	Bag 2
>270 mg/dL (>15 mmol/L)	mL/hour*	None
145 to 270 mg/dL (8 to 15 mmol/L)	mL/hour	mL/hour
70 to <145 mg/dL (4 to <8 mmol/L)	None	mL/hour
<70 mg/dL (<4 mmol/L) <i>Call clinician immediately</i>	None	mL/hour**
* These patients are receiving insulin but no glucose at this stage. ** This is a large volume of fluid and should only be continued long enough (for non-severe DKA) to raise the glucose to > 70 mg/dL (or > 4 mmol/L), then immediately decrease the rate to standard-speed rate. If BGL decreases too rapidly despite adjusting the fluid rates, insulin dose needs to be decreased. See MSF paediatric care guideline.		

 IV insulin rate
Once shock corrected, and one hour after starting IV fluid replacement, start IV insulin. If syringe pump unavailable do not give insulin IV, refer to MSF Paediatric Care guidance. See preparation of IV insulin guidance overleaf.
Units/hour
Providing insulin made up as 50 units in 50 mL (1 unit per mL) the insulin rate in mL/hour is the same as in Units/hour above.

 Prescriber		
Signature	Name & role	Date & time

Supporting notes

1. Assess the patient using a structured ABCDE approach and manage any life-threatening issues.
2. Refer to MSF Paediatric Care section 9.2 diabetic ketoacidosis for guidance on diagnostic criteria and resuscitation of patients with DKA.
3. Use this sheet to transcribe and prescribe the values from the MSF Diabetes Calculator (diabetes.msf.net). Check values before use.
4. Start by giving the fluid bolus specified by the prescriber on page 1 of this document.
5. While this is running, prepare the IV fluids for the two-bag method per *Preparation of IV fluids for two-bag method* section below.
6. Once the fluid bolus is complete, unless further boluses are clinically indicated, proceed to start the fluid replacement at the rates specified by the prescriber on page 1 of this document according to the blood glucose.
7. Once shock corrected, and one hour after starting IV fluid replacement, start IV insulin.

Preparation of IV fluids for two-bag method

Make IV fluid bag #1:			
• Add 40 mmol/L KCl (use KCl 15%, 150 mg/mL = 2 mmol/mL (10 mL ampoule))			
RL (or NaCl 0.9%)	Volume of KCl 15% to add		Final IV fluid bag #1
1000 mL	20 mL (2 ampoules)		RL + 40 mmol/L KCl (or NaCl 0.9% + 40 mmol/L KCl)
500 mL	10 mL (1 ampoule)		
Make IV fluid bag #2:			
1. Add glucose (dextrose) 50% (G50%) to a bag of Ringer lactate (RL) to make G10%-RL (or G10%- NaCl 0.9%). Follow instructions below.			
2. Add 40 mmol/L KCl.			
1. Make G10%-RL (or G10%-NaCl 0.9%)			
RL (or NaCl 0.9%)	Remove	Add	Intermediary solution
1000 mL	200 mL	G50% 200 mL	1000 mL G10%-RL (or G10%-NaCl 0.9%)
500 mL	100 mL	G50% 100 mL	500 mL G10%-RL (or G10%-NaCl 0.9%)
2. Add potassium to make final IV fluid bag #2			
G10%-RL (or G10%-NaCl 0.9%)	Volume of KCl 15% to add		Final IV fluid bag #2
1000 mL	20 mL (2 ampoules)		G10%-RL + 40 mmol/L KCl (or G10%-NaCl 0.9% + 40 mmol/L KCl)
500 mL	10 mL (1 ampoule)		

* RL = Ringer lactate, G = glucose, NaCl = sodium chloride, KCl = potassium chloride

Start fluid replacement without potassium if no urine output and add potassium to fluids only once urine output confirmed.

Preparation of IV insulin

Once shock corrected, and one hour after starting IV fluid replacement, start IV insulin. Add 50 IU of soluble insulin to 49.5 mL sodium chloride 0.9% to make a total of 50 mL. Ensure insulin preparation is double checked by two qualified healthcare professionals.